EXPLICIT HECKE CHARACTERS AND MOTIVIC MODULARITY OF CM K3 SURFACES

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Allowed standard prerequisites (editorial).

- Original arithmetic geometry and number theory prerequisites. Original complete bibliography and source-local definitions retained.

1. Introduction

Let A be an abelian variety of dimension n with CM (complex multiplication) defined over a number field $k \subset \mathbb{C}$. Shimura and Taniyama [25] showed that there exist algebraic Hecke characters $\chi_1, \cdots, \chi_{2n}$ over k such that

$$L(s, A) = \prod_{i=1}^{2n} L(s, \chi_i), \tag{1.1}$$

where $L(s, A)$ is the Hasse-Weil L-function of A and $L(s, \chi_i)$ are the Hecke L-functions of χ_i. This statement is a consequence of **the main theorem of complex multiplication** for abelian varieties.

Piatetski-Shapiro and Shafarevich [21] proved an analogue of (1.1) for **K3 surfaces with complex multiplication (CM K3 surfaces, in short)** using their Kuga-Satake constructions and Shimura-Taniyama's result.

In some arithmetic problems, e.g., the calculation of Brauer groups, it is important to know the explicit description of the algebraic Hecke characters. The explicit descriptions of Hecke characters associated to CM K3 surfaces have been obtained in the following cases:

- **K3 surfaces with non-symplectic group actions.** Livné, Schütt, and Yui [14] discussed the fields of definition and the Hecke characters of K3 surfaces with non-symplectic group actions. In such a class, K3 surfaces with the minimal Picard rank are of CM type (Example 2.20 (2)). They showed that the fields of definition of these K3 surfaces are cyclotomic fields over $\mathbb{Q}$ and the associated algebraic Hecke characters are given by **Jacobi sums**.
- **Kummer surfaces associated to the self-product of a CM elliptic curve.** Deuring [6] constructed the algebraic Hecke characters for CM elliptic curves defined over number fields, and proved the equality (1.1) for $n = 1$. Shioda and Inose [26] showed that the Hecke characters of the Kummer surface associated to the self-product of a CM elliptic curve are given by the square of the Hecke character of the CM elliptic curve.
- **Kummer surfaces associated to the product of two CM elliptic curves.** Rizov [23] proved the main theorem of complex multiplication for the products of two elliptic curves. Using the Kummer construction, one can compute the algebraic Hecke characters of the associated Kummer surfaces.
- **Certain K3 surfaces with Picard rank 16.** Costa, Elsenhans, Jahnel, and Voight [3] introduced three explicit K3 surfaces of Picard rank 16, realized as double planes. They conjectured that these surfaces have CM by CM fields obtained by adjoining $\sqrt{-1}$ to certain totally real cubic fields. Under the assumption that the surfaces are of **CM type**, they proved that the 6-dimensional Galois representations associated with their transcendental parts are isomorphic to representations induced from algebraic Hecke

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characters over the corresponding CM fields. In particular, after restriction to these CM fields, the representations decompose as direct sums of algebraic Hecke characters and their conjugates.

Remark 1.1. *Note that K3 surfaces in the above cases do not have the Picard rank 8 or 14 as shown in* [14, Theorem 1 and 2] *and* [9, Corollary 2.3, Proposition 2.4, and Corollary 2.6]*. But Taelman* [29] *proved that for any CM field E over $\mathbb{Q}$ of degree at most 20, there exists a K3 surface over $\mathbb{C}$ with complex multiplication by E. The existence of CM K3 surfaces with Picard rank 8 or 14 follows from* [29]*.*

The contribution of this paper is to provide a formula (Theorem 1.3) for the algebraic Hecke characters of CM K3 surfaces over number fields, including those with Picard rank 8 and 14. Moreover, this formula includes the following new examples:

- **Kummer surfaces associated to a simple CM abelian surface (Proposition 4.7).** This result is independent of Shioda-Inose [26] and Rizov [23]. By applying the main theorem of complex multiplication for simple abelian surfaces, one can determine the Hecke characters of the associated Kummer surfaces by a method analogous to those of Shioda-Inose and Rizov, and show that they coincide with those given in Theorem 1.3.

Remark 1.2. *Although Theorem 1.3 does not prove that the three K3 surfaces in* [3] *are of CM type, it shows that, under this assumption, their transcendental Galois representations are governed by algebraic Hecke characters in the sense of this paper. In this way, their result* [3, Theorem 1.2(a)]*are recovered by our general construction once the CM type assumption is imposed.*

1.1. **Main theorem.** Let k be a number field, and let $\mathbb{A}_k^\times$ denote the idele group of k. Fix a rational prime l and an isomorphism of fields $\overline{\mathbb{Q}_l} \cong \mathbb{C}$. By class field theory, there exists a one-to-one correspondence (see Theorem 2.12):

$$\{1^{\smile}\text{dimensional locally algebraic representation}\} \leftrightarrow \{\text{algebraic Hecke character}\}$$
$$(\rho_{l,\chi} : \mathrm{Gal}(\overline{k}/k) \to \overline{\mathbb{Q}_l}^\times) \leftrightarrow (\chi : \mathbb{A}_k^\times \to \mathbb{C}^\times).$$

An n-dimensional semisimple l-adic abelian representation ρ_l is said to be **modular** if there exist algebraic Hecke characters $\chi_1, \cdots, \chi_n$ such that $\rho_l \cong \rho_{l,\chi_1} \oplus \cdots \oplus \rho_{l,\chi_n}$.

Let k be a number field and let $\overline{k}$ be an algebraic closure of k. Let X be a K3 surface over k, i.e., a smooth projective surface over k such that $H^1(X, \mathscr{O}_X) = 0$ and $\Omega^2_{X/k} \cong \mathscr{O}_X$. Since the Galois group $\mathrm{Gal}(\overline{k}/k)$ acts on $X_{\overline{k}} := X \times_{\mathrm{Spec}(k)} \mathrm{Spec}(\overline{k})$, we obtain the 22-dimensional l-adic Galois representation

$$\rho_l : \mathrm{Gal}(\overline{k}/k) \to GL(H^2(X_{\overline{k}}, \mathbb{Q}_l(1))). \tag{1.2}$$

Fix an embedding $\overline{k} \hookrightarrow \mathbb{C}$. Then there are isomorphisms

$$H^2(X_{\overline{k}}, \mathbb{Q}_l(1)) \cong H^2(X_{\mathbb{C}}, \mathbb{Q}(1)) \otimes \mathbb{Q}_l \cong (\mathrm{NS}(X_{\mathbb{C}}) \otimes \mathbb{Q}_l) \oplus (T(X_{\mathbb{C}}) \otimes \mathbb{Q}_l),$$

where $T(X_{\mathbb{C}}) := \mathrm{NS}(X_{\mathbb{C}})^\perp \subset H^2(X_{\mathbb{C}}, \mathbb{Z}(1))$. Since $\mathrm{NS}(X_{\mathbb{C}}) \cong \mathrm{NS}(X_{\overline{k}})$, the group $\mathrm{Gal}(\overline{k}/k)$ acts on $\mathrm{NS}(X_{\mathbb{C}}) \otimes \mathbb{Q}_l$. Hence we obtain a $(22 - \rho(X_{\overline{k}}))$-dimensional Galois representation

$$\mathrm{Gal}(\overline{k}/k) \to GL(T_l(X_{\overline{k}}) \otimes \mathbb{Q}_l), \tag{1.3}$$

where $T_l(X_{\overline{k}}) := T(X_{\mathbb{C}}) \otimes \mathbb{Z}_l$. The $\mathbb{Z}_l$-module $T_l(X_{\overline{k}})$ is called the **l-adic transcendental lattice** of X.

We obtain the following new result.

Theorem 1.3. *Let X be a K3 surface defined over a number field $k \subset \mathbb{C}$. Suppose that X has CM over k by a CM field E, i.e., $X_{\mathbb{C}}$ is of CM type and $k \supset E$. Then there exists a unique*

locally constant homomorphism $u : \mathbb{A}_k^\times \to E^\times \subset \mathbb{A}_E^\times$ *satisfying the following properties (i)-(iii):*

(i) Fix an embedding $\tau : E \to \mathbb{C}$*. The function*

$$\chi_\tau : \mathbb{A}_k^\times \xrightarrow{u(-)\cdot\overline{\mathrm{Nm}_{k/E}(-)}/\mathrm{Nm}_{k/E}(-)} \mathbb{A}_E^\times \xrightarrow{\tau-\text{projection}} \mathbb{C}^\times$$

is an algebraic Hecke character, where $\mathrm{Nm}_{k/E} : \mathbb{A}_k^\times \to \mathbb{A}_E^\times$ *denotes the norm map for the extension* k/E*, and* $\overline{(-)} : \mathbb{A}_E^\times \to \mathbb{A}_E^\times$ *denotes the complex conjugation.*

(ii) Let

$$\rho_l : G_k \to GL(T_l(X_{\overline{k}}) \otimes \mathbb{Q}_l)$$

be the $(22 - \rho(X_{\overline{k}}))$*-dimensional* l*-adic representation. Then* ρ_l *is diagonalized by the algebraic Hecke characters* χ_τ*, where* τ *runs over* $\mathrm{Hom}(E, \mathbb{C})$*.*

(iii) Fix an embedding $\tau \in \mathrm{Hom}(E, \mathbb{C})$ *and a prime ideal* $\mathfrak{p} \subset k$ *with* $(l, \mathfrak{p}) = 1$*. Then the following conditions are equivalent:*

(a) The l*-adic representation* ρ_l *is unramified at* $\mathfrak{p}$*, i.e., if* $I_\mathfrak{p}$ *denotes the inertia group of* $\mathfrak{p}$*, then* $\rho_l(I_\mathfrak{p}) = \{id\}$*.*

(b) The Hecke character χ_τ *is unramified at* $\mathfrak{p}$*, i.e., if* $\mathscr{O}_\mathfrak{p}$ *is the ring of integers of* $k_\mathfrak{p}$*, then* $\chi_\tau(\mathscr{O}_\mathfrak{p}^\times) = 1$*.*

Remark 1.4. *If* k *does not contain* E*, Theorem 1.3 can be extended to a K3 surface* X *over* kE*, and Theorem 1.3 is valid for K3 surface* X *over* kE*.*

By Theorem 1.3, we obtain the following equality

$$L(s, T_l(X_{\overline{k}}) \otimes \mathbb{Q}_l) = \prod_{\tau \in \mathrm{Hom}(E,\mathbb{C})} L(s, \chi_\tau). \tag{1.4}$$

Here $L(s, T_l(X_{\overline{k}}) \otimes \mathbb{Q}_l)$ denotes the L-function of ρ_l (see [30, Section 2]). We call the equality (1.4) the **Motivic modularity** of X.

A key point in the proof of Theorem 1.3 is to construct algebraic Hecke characters χ_τ satisfying condition (i) of Theorem 1.3. The fact that the characters χ_τ satisfy the property (ii) follows from the semisimplicity of ρ_l. These are applications of the main theorem of complex multiplication for K3 surfaces by Rizov [23].

On the other hand, the semisimplicity also follows from the theory of absolute Hodge cycles by Deligne [5]; Indeed, let X be a K3 surface with CM over k by a CM field E. The Hodge cycles

$$E := \mathrm{End}_{Hdg} T(X_\mathbb{C})_\mathbb{Q} \subset H^4(X_\mathbb{C} \times X_\mathbb{C}, \mathbb{Q})(2)$$

are absolute Hodge cycles (as a consequence of Deligne's Principle B [5, I, Theorem 2.12] and the Kuga-Satake construction [4, Proposition 6.5]). Hence the action of $\mathrm{Gal}(\overline{k}/k)$ on $T(X_\mathbb{C})_{\mathbb{A}_f}$ commutes with the action of E. It follows that the image of the Galois representation $\rho : \mathrm{Gal}(\overline{k}/k) \to GL(T(X_\mathbb{C})_{\mathbb{A}_f})$ is contained in $\mathbb{A}_{E,f}^\times$. Since $T(X_\mathbb{C})_{\mathbb{A}_f}$ is a rank-1 $\mathbb{A}_{E,f}$-module, it follows that ρ is semisimple.

Remark 1.5. *In general, Zarhin and Skorobogatov* [28] *proved that the Galois representation (1.2) is semisimple for any K3 surface. Their result is based on the semisimplicity of the Galois representations for abelian varieties due to Faltings* [8]*.*

In the final section, we compute, as an example of Theorem 1.3, the algebraic Hecke characters in Theorem 1.3 for Kummer surfaces associated with CM simple abelian surfaces. The main theorem of complex multiplication for simple abelian surfaces describes the Galois action on the **Tate module** (Milne [17]). Using this result, we compute the Galois action on the **transcendental part** of a CM simple abelian surface (see Proposition 4.4). Finally, we show that the Hecke characters obtained from Proposition 4.4 coincide with those given in Theorem 1.3.

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2. Preliminaries

2.1. **Abelian l-adic representations.** For a rational prime l, let $\mathbb{Q}_l$ be the l-adic field, and let $\overline{\mathbb{Q}_l}$ be an algebraic closure of $\mathbb{Q}_l$.

Definition 2.1. *(Abelian l-adic representations)*

*Let k be a number field. An **l-adic Galois representation** is a finite-dimensional $\overline{\mathbb{Q}_l}$-continuous representation of the Galois group $\mathrm{Gal}(\overline{k}/k)$. An **abelian l-adic representation** is an l-adic Galois representation whose image is commutative.*

Example 2.2. Let k be a number field, let $\overline{k}$ be an algebraic closure, and let X be a smooth projective variety defined over k. Let $X_{\overline{k}} := X \times_{\mathrm{Spec}(k)} \mathrm{Spec}(\overline{k})$. Then the i-th étale cohomology $H^i(X_{\overline{k}}, \mathbb{Q}_l)$ is a finite-dimensional $\mathbb{Q}_l$-vector space. If $\sigma \in \mathrm{Gal}(\overline{k}/k)$, it induces the automorphism of $X_{\overline{k}}$. Hence we obtain the l-adic Galois representation

$$\rho_l : \mathrm{Gal}(\overline{k}/k) \to GL(H^i(X_{\overline{k}}, \mathbb{Q}_l)). \tag{2.1}$$

Now we recall the main statements of global class field theory. Let k be a number field, let $\mathbb{A}_k^\times$ be the idele group, and let k^{ab} be the maximal abelian extension of k. Let $[-,k] : \mathbb{A}_k^\times \to \mathrm{Gal}(k^{ab}/k)$ be the reciprocity map for k. The **Artin map** $\mathrm{art}_k : \mathbb{A}_k^\times \to \mathrm{Gal}(k^{ab}/k)$ is defined by $\mathrm{art}_k := [-,k]^{-1}$.

Theorem 2.3. [27, Chapter II, Theorem 3.5] *Let k be a number field. The Artin map art_k has the following properties :*

(i) The Artin map is surjective and $k^\times$ is contained in the kernel of art_k.

(ii) For any finite abelian extension K/k, the Artin map is compatible with the norm map $\mathrm{Nm}_{K/k} : \mathbb{A}_K^\times \to \mathbb{A}_k^\times$, i.e., for all $x \in \mathbb{A}_K^\times$ the following diagram

$$\begin{array}{ccc} \mathbb{A}_K^\times & \xrightarrow{\mathrm{art}_K(x)} & \mathrm{Gal}(K^{ab}/K) \\ \downarrow{\scriptstyle \mathrm{Nm}_{K/k}} & & \downarrow \\ \mathbb{A}_k^\times & \xrightarrow[\mathrm{art}_k(\mathrm{Nm}_{K/k}(x))]{} & \mathrm{Gal}(k^{ab}/k) \end{array}$$

is commutative.

(iii) Let $\mathfrak{p}$ be a prime ideal of k, let $I_\mathfrak{p}^{ab} \subset \mathrm{Gal}(k^{ab}/k)$ be an inertia group of $\mathfrak{p}$ for the extension k^{ab}/k, let $\pi_\mathfrak{p}$ be a uniformizer at $\mathfrak{p}$, and let K/k be any abelian extension that is unramified at $\mathfrak{p}$. Then $\mathrm{art}_k(\pi_\mathfrak{p})^{-1}|_K$ is the Frobenius for K/k at $\mathfrak{p}$ and $\mathrm{art}_k(\mathscr{O}_\mathfrak{p}^\times) = I_\mathfrak{p}^{ab}$ where $\mathscr{O}_\mathfrak{p}$ is the ring of integers of $k_\mathfrak{p}$.

Let k be a number field, and denote by Σ_k the set of all finite places of k. Let $v \in \sum_k$ be a finite place of k, and let $\mathfrak{p}$ be the prime ideal corresponding to v. Put $q := |\mathscr{O}_k/\mathfrak{p}|$. Then there exists the following exact sequence:

$$1 \to I_\mathfrak{p} \to \mathrm{Gal}(\overline{k_\mathfrak{p}}/k_\mathfrak{p}) \to \mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) \to 1, \tag{2.2}$$

where $I_\mathfrak{p}$ is the inertia group at $\mathfrak{p}$. The group $\mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q)$ is topologically generated by the **geometric Frobenius** $x \mapsto x^{q^{-1}}$. By the exact sequence (2.2), there exists an element $\mathrm{Frob}_\mathfrak{p} \in$

$\mathrm{Gal}(\overline{k}_{\mathfrak{p}}/k_{\mathfrak{p}})$ which induces the geometric Frobenius on $\overline{\mathbb{F}_q}$, up to the action of $I_{\mathfrak{p}}$. This element $\mathrm{Frob}_{\mathfrak{p}}$ is called the **(geometric) Frobenius element** at $\mathfrak{p}$.

Definition 2.4. *Let $v \in \sum_k$ and let $\mathfrak{p}$ be the prime ideal associated to v. An l-adic representation ρ_l is said to be **unramified** at v if $\rho_l(I_{\mathfrak{p}}) = \{id\}$.*

Proposition 2.5. *Let X be a smooth projective surface defined over a number field k. Suppose that the l-adic representation ρ_l is abelian, i.e., the image is abelian. Then ρ_l induces a Galois representation*

$$\rho_l^{ab} : \mathrm{Gal}(k^{ab}/k) \to GL(H^2_{et}(X_{\overline{k}}, \mathbb{Q}_l)).$$

Moreover, ρ_l is unramified at v if and only if $\rho_l^{ab}(I_{\mathfrak{p}}^{ab}) = \{id\}$ where $\mathfrak{p}$ is the prime of k associated to v.

Proof. Let q be the norm of $\mathfrak{p}$. Since finite extensions of $\mathbb{F}_q$ are cyclic, we have $\overline{\mathbb{F}_q} = \mathbb{F}_q^{ab}$. Consider the following commutative diagram:

$$\begin{array}{ccccc} I_{\mathfrak{p}} & \longrightarrow & \mathrm{Gal}(\overline{k_{\mathfrak{p}}}/k_{\mathfrak{p}}) & \longrightarrow & \mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) \\ \downarrow & & \downarrow{\scriptstyle res} & & \downarrow{\scriptstyle id} \\ I_{\mathfrak{p}}^{ab} & \longrightarrow & \mathrm{Gal}(k_{\mathfrak{p}}^{ab}/k_{\mathfrak{p}}) & \longrightarrow & \mathrm{Gal}(\mathbb{F}_q^{ab}/\mathbb{F}_q). \end{array}$$

Suppose that $\rho_l(I_{\mathfrak{p}}) = \{id\}$. Let $\sigma \in I_{\mathfrak{p}}^{ab}$. Then there exists an element $\sigma' \in \mathrm{Gal}(\overline{k}/k)$ such that $\sigma'|_{k^{ab}} = \sigma$. This implies that $\rho_l^{ab}(\sigma) = \rho_l(\sigma')$. On the other hand, since $I_{\mathfrak{p}}^{ab}$ is the kernel of $\mathrm{Gal}(k_{\mathfrak{p}}^{ab}/k_{\mathfrak{p}}) \to \mathrm{Gal}(\mathbb{F}_q^{ab}/\mathbb{F}_q)$, it follows from the above diagram that $\sigma' \in I_{\mathfrak{p}}$. Hence we have $\rho_l^{ab}(\sigma) = \rho_l(\sigma') = id$.
The converse is clear. □

If an l-adic representation $\rho_l : \mathrm{Gal}(\overline{k}/k) \to GL(V)$ is unramified at v, then the action of $\rho_l(\mathrm{Frob}_{\mathfrak{p}})$ on V is well-defined. Then we denote the characteristic polynomial of $\rho_l(\mathrm{Frob}_{\mathfrak{p}})$ by $P_{v,\rho_l}(T)$.

Definition 2.6. *(Rational representations)*
*An l-adic representation ρ_l is said to be **rational** if there exists a finite subset $S \subset \sum_k$ such that:*
(i) Any element of $\sum_k \setminus S$ is unramified for ρ_l.
(ii) If $v \notin S$, the coefficients of $P_{v,\rho_l}(T)$ lie in $\mathbb{Q}$.

Remark 2.7. *Dwork* [7] *proved that the Galois representation (2.1) is rational. This is a part of the Weil conjecture.*

Definition 2.8. *(Hecke characters)*
*Let k be a number field. A continuous homomorphism $\chi : \mathbb{A}_k^\times \to \mathbb{C}^\times$ is called a **Hecke character** if χ is trivial on $k^\times$.*

Definition 2.9. *Let k be a number field. Let $v \in \sum_k$ be a finite place, and let $\mathfrak{p}$ be the prime ideal associated with v. A Hecke character $\chi : \mathbb{A}_k^\times \to \mathbb{C}^\times$ is said to be **unramified** at v if it is trivial on $\mathscr{O}_{\mathfrak{p}}^\times \subset \mathbb{A}_k^\times$.*

We recall that there exists a correspondence between the set of one-dimensional locally algebraic Galois representations and the set of algebraic Hecke characters. Let k be a number field, and let v be an infinite place. We denote by $k_v^{\times 0}$ the connected component of the identity of k_v.

Definition 2.10. *(Algebraic Hecke characters)*
*Let k be a number field. A Hecke character $\chi : \mathbb{A}_k^\times \to \mathbb{C}^\times$ is said to be **algebraic** if, for any infinite place v, there exist integers $(a_\tau)_{\tau \in \mathrm{Hom}_{\mathbb{R}}(k_v, \mathbb{C})}$ such that*

$$\chi_v(x_v) = \prod_{\tau \in \mathrm{Hom}_{\mathbb{R}}(k_v, \mathbb{C})} \tau(x_v)^{-a_\tau}, \ (x_v \in k_v^{\times 0}),$$

where $\chi_v : k_v^\times \to \mathbb{C}^\times$ is the v-component of χ.

Definition 2.11. *(Locally algebraic representations)*

Let k be a number field, and let $\rho_l : \mathrm{Gal}(k^{ab}/k) \to GL(V)$ be an l-adic abelian representation.

*(i) Let $v|l$. Put $\rho_v := \rho_l|_{\mathrm{Gal}(\overline{k_v}/k_v)}$, and let $T := \mathrm{Res}_{k_v/\mathbb{Q}_l}(\mathbb{G}_{m,k_v})$. The representation ρ_l is said to be **locally algebraic at** v if there exists an algebraic morphism $r : T \to GL(V)$ such that the following diagram commutes on a neighborhood of 1 in $k_v^\times$:*

$$\begin{array}{ccc} \mathrm{Gal}(k_v^{ab}) & \xrightarrow{\rho_v} & GL(V) \\ {\scriptstyle \mathrm{art}_k}\uparrow & \nearrow_{r} & \\ k_v^\times. & & \end{array}$$

*(ii) The l-adic abelian representation ρ_l is said to be **locally algebraic** if it is locally algebraic at all places $v|l$.*

Theorem 2.12. [24, Chapter III-1-Proposition 2] *Let l be a prime. Fix an isomorphism of fields $\iota : \overline{\mathbb{Q}_l} \to \mathbb{C}$. Let k be a number field. Then there exists a one-to-one correspondence*

$$\{1\breve{}\text{dimensional locally algebraic representation}\} \leftrightarrow \{\text{algebraic Hecke character}\}$$
$$\rho_l \leftrightarrow \chi.$$

Under this correspondence, we have $\chi_v = \iota \circ \rho_{l,v} \circ \mathrm{art}_k$ for all finite places $v \nmid l$.

2.2. **Mumford-Tate groups.** Deligne gave a representation-theoretic definition and characterization of Hodge structures in [5]. We now recall them. Let V be a rational Hodge structure of weight n. In other words, there exists a decomposition $V_{\mathbb{C}} = \bigoplus_{p+q=n} V^{p,q}$ with $\overline{V^{p,q}} = V^{q,p}$. This gives a real representation

$$\rho : \mathbb{C}^\times \to GL(V_{\mathbb{R}}), z \mapsto \rho(z) : v \mapsto (z^{-p}\overline{z}^{-q})v$$

for $v \in V^{p,q}$. In particular, for $t \in \mathbb{R}^\times$, we have $\rho(t) = t^{-n} : v \mapsto t^{-n}v$. Conversely, if V is a finite-dimensional $\mathbb{Q}$-vector space and $\rho : \mathbb{C}^\times \to GL(V_{\mathbb{R}})$ is a real representation such that $\rho(t) = t^{-n}$ for all $t \in \mathbb{R}^\times$, then the $\mathbb{C}$-subspaces

$$V^{p,q} := \{v \in V_{\mathbb{C}} |\ \rho(z)(v) = z^{-p}\overline{z}^{-q}v, \text{for all } z \in \mathbb{C}^\times\},\ p+q=n$$

define a rational Hodge structure of weight n on V. Hence we obtain a one-to-one correspondence:

$$\{\text{rational Hodge structure of weight } n\} \leftrightarrow \{\text{real representation } \mathbb{C}^\times \to GL(V_{\mathbb{R}})$$
$$\text{such that } t \mapsto t^{-n} \cdot (-) \text{ on } \mathbb{R}^\times\}.$$

On the other hand, a real representation $\mathbb{C}^\times \to GL(V_{\mathbb{R}})$ can be regarded as a representation of the Deligne torus $\mathbb{S} := \mathrm{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}})$. This is an $\mathbb{R}$ algebraic group defined by

$$\{\mathbb{R}\breve{}\text{algebras}\} \ni A \mapsto \mathbb{S}(A) := (A \otimes_{\mathbb{R}} \mathbb{C})^\times \in \{\text{groups}\}.$$

Hence, a rational Hodge structure on V of weight n corresponds to an algebraic $\mathbb{R}$-morphism $\rho : \mathbb{S} \to GL(V_{\mathbb{R}})$ such that $\rho(t) = t^{-n}$ for all $t \in \mathbb{R}^\times$.

Definition 2.13. *(Mumford-Tate groups)*

*Let V be a rational Hodge structure, and let $\rho : \mathbb{S} \to GL(V_{\mathbb{R}})$ be the algebraic $\mathbb{R}$-representation associated with V. The **Mumford-Tate group** $\mathrm{MT}(V)$ of V is the smallest $\mathbb{Q}$-algebraic subgroup of $GL(V)$ such that $\rho(\mathbb{S}(\mathbb{R})) \subset \mathrm{MT}(V)(\mathbb{R})$.*

Example 2.14. (i) Let X be an algebraic K3 surface over $\mathbb{C}$. Consider the second cohomology $H^2(X,\mathbb{Z})$. This is a Hodge structure of weight 2, hence the Tate twist $H^2(X,\mathbb{Z})(1)$ is of weight 0. By the Lefschetz $(1,1)$-theorem (see [10, Chapter III, Proposition 3.3.2]), the Néron-Severi group $\mathrm{NS}(X)$ is isomorphic to $H^2(X,\mathbb{Z})(1) \cap H^{1,1}(X)$. Hence the orthogonal complement $T(X)$ is a sub-Hodge structure of weight 0.

(ii) Let A be an abelian surface over $\mathbb{C}$. Then the second cohomology $H^2(A,\mathbb{Z})$ is a Hodge structure of weight 2, hence the Tate twist $H^2(A,\mathbb{Z})(1)$ is of weight 0. By the Lefschetz $(1,1)$-theorem, $\mathrm{NS}(A) \cong H^2(A,\mathbb{Z})(1) \cap H^{1,1}(A)$. Hence the orthogonal complement $T(A)$ is a sub-Hodge structure of weight 0.

2.3. **K3 surfaces with CM.**

To explain CM, we first define CM fields.

Definition 2.15. *(CM fields) Let K be a number field.*
*(i) K is said to be **totally real** if $\sigma(K) \subset \mathbb{R}$ for all inclusions $\sigma : K \to \mathbb{C}$.*
*(ii) K is said to be **totally imaginary** if $\sigma(K) \not\subset \mathbb{R}$ for all inclusions $\sigma : K \to \mathbb{C}$.*
*(iii) A **CM field** is a totally imaginary quadratic extension of a totally real field.*

Example 2.16. (1) An imaginary quadratic field is a CM field. In fact, $\mathbb{Q}$ is totally real, and the imaginary quadratic extension is not contained in $\mathbb{R}$.

(2) Let p be a prime $\neq 2$, and let ζ_p be a p-th root of unity. Consider the field extensions $\mathbb{Q} \subset \mathbb{Q}(\zeta_p+\zeta_p^{-1}) \subset \mathbb{Q}(\zeta_p)$. Since $\zeta_p^2-(\zeta_p+\zeta_p^{-1})\zeta_p+1=0$, the extension $\mathbb{Q}(\zeta_p+\zeta_p^{-1}) \subset \mathbb{Q}(\zeta_p)$ has degree 2. If $\sigma : \mathbb{Q}(\zeta_p) \to \mathbb{C}$ is an inclusion, there exists an integer n with $\sigma(\zeta_p) = \zeta_p^n, (n,p)=1$. This shows that $\mathbb{Q}(\zeta_p)$ is totally imaginary. Finally, the field $\mathbb{Q}(\zeta_p+\zeta_p^{-1})$ is totally real because

$$\sigma(\zeta_p+\zeta_p^{-1}) = \zeta_p^n + \zeta_p^{-n} = 2\cos\left(\frac{2n\pi}{p}\right) \in \mathbb{R}.$$

Hence $\mathbb{Q}(\zeta_p)$ is a CM field.

Let X be an algebraic K3 surface over $\mathbb{C}$. Let $E_X := \mathrm{End}_{\mathrm{Hdg}}(T(X)_{\mathbb{Q}})$. If ω is a nowhere vanishing holomorphic 2-form on X, then we have a homomorphism of $\mathbb{Q}$-algebras:

$$\sigma_X : E_X \to \mathrm{End}_{\mathbb{C}}(\mathbb{C}\omega) \cong \mathbb{C}. \tag{2.3}$$

Note that this morphism is independent of the choice of ω.

Proposition 2.17. *Let X be a complex algebraic K3 surface. Then the homomorphism σ_X of (2.3) is injective, and E_X is a number field.*

Proof. See [11, Chapter III, Corollary 3.6]. □

Definition 2.18. *(CM K3 surfaces over $\mathbb{C}$)*
*Let X be an algebraic K3 surface over $\mathbb{C}$. We say that X has **CM** over $\mathbb{C}$ if the following conditions hold:*
(i) E_X is a CM field.
(ii) $\dim_{E_X} T(X)_{\mathbb{Q}} = 1$.

By condition (ii), we have $[E_X : \mathbb{Q}] = \dim_{\mathbb{Q}} T(X)_{\mathbb{Q}}$.

Remark 2.19. *Valloni* [31] *proved that the **reflex field** of the rational Hodge structure $T(X)_{\mathbb{Q}}$ is $\sigma_X(E_X)$.*

Example 2.20. (1) If X is a singular K3 surface, i.e., it has the maximal Picard rank 20, then X has CM over $\mathbb{C}$. Then the reflex field is an imaginary quadratic field. These considerations are given in [11, Chapter 3, Remark 3.10].

(2) Let X be a K3 surface over $\mathbb{C}$. Suppose that there exists an automorphism $\sigma \in \mathrm{Aut}(X)$ of order N such that the transcendental rank $\mathrm{rank}_{\mathbb{Z}} T(X)$ is equal to $\varphi(N)$, where φ is Euler's function. Then X has CM over $\mathbb{C}$, and the reflex field is the cyclotomic field $\mathbb{Q}(\zeta_N)$, where ζ_N is a primitive N-th root of unity.

(3) Let A be an abelian surface over $\mathbb{C}$, and let X be the Kummer surface associated to A. Then X has CM over $\mathbb{C}$ if and only if A has CM over $\mathbb{C}$ (see [12, Lemma 9.3]).

Definition 2.21. *Let X be a K3 surface over a number field $k \subset \mathbb{C}$. Then we say that X has CM over k if*

(i) the scalar extension $X_{\mathbb{C}} := X \times_{\mathrm{Spec}(k)} \mathrm{Spec}(\mathbb{C})$ has CM over $\mathbb{C}$,

(ii) $k \supset \sigma_{X_{\mathbb{C}}}(E_{X_{\mathbb{C}}})$.

Remark 2.22. *Mukai, Nikulin, and Buskin([18], [20], and [1]) proved that the Hodge conjecture for the self-product of a CM K3 surface is true (see* [1, Corollary of Theorem 1.1]*). Using this, Valloni* [31] *gave a condition for X to have complex multiplication over k, analogous to the corresponding condition for abelian varieties.*

Finally, we introduce a fundamental fact for CM K3 surfaces. Let X be a K3 surface with CM over $\mathbb{C}$, and let $G := \mathrm{MT}(T(X)_{\mathbb{Q}})$. Zarhin proved the following (see [33, Theorem 2.3.1]):

Theorem 2.23. *The algebraic group G is isomorphic, as a $\mathbb{Q}$-algebraic group, to the kernel of*

$$\mathrm{Nm} : \mathrm{Res}_{E/\mathbb{Q}}\mathbb{G}_{m,E} \to \mathrm{Res}_{F/\mathbb{Q}}\mathbb{G}_{m,F},$$

where $E = \sigma_X(E_X)$ and F is the maximal totally real subfield of E.

Hence the rational points of G are given by

$$G(\mathbb{Q}) = \{x \in E^{\times} | x\overline{x} = 1\},$$

where $\overline{(-)}$ denotes the complex conjugation $E \to E$.

3. The proof of Theorem 1.3

3.1. **The main theorem of complex multiplication by Rizov.** Suppose that X has CM over a number field k. Let $\mathbb{A}_f := (\prod_l \mathbb{Z}_l) \otimes_{\mathbb{Z}} \mathbb{Q}$ be the ring of finite adeles of $\mathbb{Q}$. Fix an embedding $\overline{k} \to \mathbb{C}$. Then we have an isomorphism

$$H^2(X_{\overline{k}}, \mathbb{Z}_l(1)) \cong H^2(X_{\mathbb{C}}, \mathbb{Z}(1)) \otimes_{\mathbb{Z}} \mathbb{Z}_l.$$

Hence

$$(\prod_l H^2(X_{\overline{k}}, \mathbb{Z}_l(1)) \otimes_{\mathbb{Z}} \mathbb{Q} \cong H^2(X_{\mathbb{C}}, \mathbb{Z}(1)) \otimes_{\mathbb{Z}} \mathbb{A}_f,$$

and

$$(\prod_l T_l(X_{\overline{k}})) \otimes_{\mathbb{Z}} \mathbb{Q} \cong T(X_{\mathbb{C}}) \otimes_{\mathbb{Z}} \mathbb{A}_f.$$

Thus we obtain a Galois representation $\rho : \mathrm{Gal}(\overline{k}/k) \to GL(T(X_{\mathbb{C}})_{\mathbb{Q}})(\mathbb{A}_f)$ whose image is contained in $G(\mathbb{A}_f)$

To construct Hecke characters $\mathbb{A}_k^{\times} \to \mathbb{C}^{\times}$, we first recall the main theorem of CM for $X_{\mathbb{C}}$ by Rizov[23]. This gives a description of ρ via class field theory. Let E be the reflex field of $X_{\mathbb{C}}$. Since X has CM over k, we have $E \subset k$ by definition 2.21. Since E is a CM field, complex conjugation $\overline{(-)}$ is an automorphism of E. Hence it induces an involution $\overline{(-)} : \mathbb{A}_E^{\times} \to \mathbb{A}_E^{\times}$.

The main theorem of complex multiplication was proved by Rizov [23]. The following statement follows from [12, Theorem 2.1]. The proof is given in [15, Corollary 4.4].

Theorem 3.1. *The action of $\mathrm{Gal}(\overline{k}/k)$ on $T(X_{\mathbb{C}}) \otimes \mathbb{A}_f$ factors through a continuous homomorphism*

$$\rho^{ab} : \mathrm{Gal}(k^{ab}/k) \to G(\mathbb{A}_f) = \{x \in \mathbb{A}_{E,f}^{\times} | x\overline{x} = 1\},$$

and the following diagram commutes:

$$\begin{array}{ccccc}
\mathbb{A}_k^{\times} & \xrightarrow{\mathrm{art}_k} & \mathrm{Gal}(k^{ab}/k) & \xrightarrow{\rho^{ab}} & G(\mathbb{A}_f) \\
\downarrow{\scriptstyle \mathrm{Nm}_{k/E}} & & & & \downarrow{\scriptstyle quotient} \\
\mathbb{A}_E^{\times} & \xrightarrow[proj]{} & \mathbb{A}_{E,f}^{\times} & \xrightarrow[\overline{(-)}/(-)]{} & G(\mathbb{A}_f)/G(\mathbb{Q}).
\end{array}$$

Remark 3.2. *By Theorem 3.1, the image of ρ is contained in $\mathbb{A}_{E,f}^{\times}$. Since $T(X_{\mathbb{C}})_{\mathbb{A}_f}$ is a free $\mathbb{A}_{E,f}$-module of rank one, it follows that ρ is semisimple.*

Remark 3.3. *(Valloni* [32]*)*

In the situation of Theorem 3.1, let $t \in \mathbb{A}_k^{\times}$ and put $s := \mathrm{Nm}_{k/E}(t) \in \mathbb{A}_{E,f}^{\times}$. Then there exists a unique $u \in G(\mathbb{Q})$ such that

$$\rho^{ab}(\mathrm{art}_k(t)) = u \cdot \overline{s}/s \in G(\mathbb{A}_f).$$

The element $\overline{s}/s$ has the following property:

Proposition 3.4. *Let v be a finite place of k, and let $t \in \mathrm{Im}(\mathscr{O}_{p_v}^{\times} \to \mathbb{A}_k^{\times})$. Put $s := \mathrm{Nm}_{k/E}(t)$. Under the isomorphism $\prod_l T_l(X_{\overline{k}}) \otimes \mathbb{Q} \cong T(X_{\mathbb{C}}) \otimes \mathbb{A}_f$, for $(l, \mathfrak{p}_v) = 1$, the l-th component of $\overline{s}/s \in GL(T(X_{\mathbb{C}}))(\mathbb{A}_f)$ is equal to 1.*

Proof. The l-th component of $\overline{s}/s$ is given by $\prod_{w|l}(\overline{s}/s)_w$. On the other hand, by the definition of the norm, the w-th component of $s \in \mathbb{A}_{E,f}^{\times}$ is given by

$$\prod_{v'|w} N_{k_{v'}/E_w}(t_{v'}).$$

Fix $w|l$. If $v|w$, then $v|l$, which is a contradiction. Hence, by assumption, we have $s_w = 1$. Similarly, $\overline{s}_w = 1$. Since w was arbitrary finite place of E with $w|l$, it follows that $(\overline{s}/s)_l = 1$. □

3.2. **Construction of Hecke characters.** The goal of this section is to prove Theorem 1.3. First, we prove the following lemma from algebraic number theory.

Lemma 3.5. *Let E be a CM field. Then the group*

$$G(\mathbb{Q}) = \{x \in E^{\times} |\ x\overline{x} = 1\}$$

is a discrete subgroup of $\mathbb{A}_{E,f}^{\times}$

Proof. Let F be the maximal totally real field of E. By Dirichlet's unit theorem, the group of units $\mathscr{O}_F^{\times}$ has a finite index in $\mathscr{O}_E^{\times}$ since E is a CM field. Moreover, under the natural inclusion $E \to \mathbb{A}_{E,f}^{\times}$, we have

$$\mathscr{O}_E = E \cap (\prod_v \mathscr{O}_v).$$

Since $\prod_v \mathscr{O}_v$ is open in $\mathbb{A}_{E,f}^{\times}$, it follows that $\mathscr{O}_E$ is open in E. Hence $\mathscr{O}_F$ is also open in E. On the other hand, $G(\mathbb{Q}) \cap \mathscr{O}_F = \{\pm 1\}$. Therefore, since $\mathbb{A}_{E,f}^{\times}$ is Hausdorff, it follows that $G(\mathbb{Q})$ is a discrete subgroup. □

Proposition 3.6. *Suppose that X is a K3 surface with CM over a number field $k \subset \mathbb{C}$. Let E be the reflex field. Then there exists a unique locally constant homomorphism $u : \mathbb{A}_k^{\times} \to E^{\times}$ satisfying the following condition:*
For all $t \in \mathbb{A}_k^{\times}$,

$$\rho^{ab}(\mathrm{art}_k(t)) = u(t)\overline{\mathrm{Nm}_{k/E}(t)}/\mathrm{Nm}_{k/E}(t) \in G(\mathbb{A}_f).$$

Proof. Under the notation of Remark 3.3, the correspondence $t \to u$ defines a map from $\mathbb{A}_k^{\times}$ to $G(\mathbb{Q}) = \{x \in E^{\times} |\ x\overline{x} = 1\}$, which we denote by u. The function u is continuous, since it is defined by a functional equation involving the nonzero continuous functions ρ^{ab} and $\mathrm{Nm}_{k/E}$. Next, we show that u is a group homomorphism. Let $t_1, t_2 \in \mathbb{A}_k^{\times}$. By Theorem 3.1, we have

$$\begin{aligned}
\rho^{ab}(\mathrm{art}_k(t_1)\mathrm{art}_k(t_2)) &= \rho^{ab}(\mathrm{art}_k(t_1))\rho^{ab}(\mathrm{art}_k(t_2)) \\
&= u(t_1)\overline{\mathrm{Nm}_{k/E}(t_1)}\mathrm{Nm}_{k/E}(t_1)^{-1}u(t_2)\overline{\mathrm{Nm}_{k/E}(t_2)}\mathrm{Nm}_{k/E}(t_2)^{-1} \\
&= u(t_1)u(t_2)\overline{\mathrm{Nm}_{k/E}(t_1t_2)}\mathrm{Nm}_{k/E}(t_1t_2)^{-1}.
\end{aligned}$$

On the other hand, since the Artin map $\mathrm{art}_k : \mathbb{A}_k^\times \to \mathrm{Gal}(k^{ab}/k)$ is a homomorphism, we have $\mathrm{art}_k(t_1)\mathrm{art}_k(t_2) = \mathrm{art}_k(t_1t_2)$. Hence, by uniqueness, we obtain $u(t_1t_2) = u(t_1)u(t_2)$. Therefore, u is a homomorphism from $\mathbb{A}_k^\times$ to $G(\mathbb{Q})$. □

Let $\tau : E \to \mathbb{C}$ be an inclusion. We define a character $\chi_\tau : \mathbb{A}_k^\times \to \mathbb{C}^\times$ by

$$\chi_\tau(t) := \left(u(t) \cdot \overline{\mathrm{Nm}_{k/E}(t)}/\mathrm{Nm}_{k/E}(t)\right)_\tau,$$

where $(-)_\tau$ denotes the τ-component in $\mathbb{A}_E^\times$.

Theorem 3.7. *(i) The characters χ_τ are algebraic Hecke characters of k.*
(ii) Let l be a prime, and let $\rho_l : \mathrm{Gal}(\overline{k}/k) \to GL(T_l(X_{\overline{k}}) \otimes \mathbb{Q}_l))$ be the l-adic Galois representation. Fix a prime ideal $\mathfrak{p} \subset k$ such that $(l, \mathfrak{p}) = 1$. Then the following conditions are equivalent:
(a) The l-adic representation ρ_l is unramified at $\mathfrak{p}$.
(b) The Hecke character χ_τ is unramified at $\mathfrak{p}$, i.e., if $\mathscr{O}_\mathfrak{p}$ is the ring of integers of $k_\mathfrak{p}$, then $\chi_\tau(\mathscr{O}_\mathfrak{p}^\times) = 1$.

Proof. (i): Let $t \in k^\times$. Since $k^\times$ is contained in the kernel of the Artin map by Theorem 2.3(i), we have $\mathrm{art}_k(t) = id_{k^{ab}}$. Since $t \in k$, we have $\mathrm{Nm}_{k/E}(t) = N_{k/E}(t)$. Hence we obtain

$$\rho(id_{\overline{k}}) = u(t) \cdot \overline{N_{k/E}(t)} N_{k/E}(t)^{-1}.$$

Thus we have $\chi_\tau(t) = 1$. This shows that χ_τ is a Hecke character.
(ii) By the Artin map, $\mathscr{O}_\mathfrak{p}^\times$ corresponds to $I_\mathfrak{p}$ (Theorem 2.3(iii)). Let $t \in \mathrm{Im}(\mathscr{O}_\mathfrak{p}^\times \to \mathbb{A}_k^\times)$.
$(a) \Rightarrow (b)$: Suppose that ρ_l is unramified at $\mathfrak{p}$. By Theorem 3.1, we obtain

$$\rho^{ab}(\mathrm{art}_k(t)) = u(t) \cdot \overline{\mathrm{Nm}_{k/E}(t)} \mathrm{Nm}_{k/E}(t)^{-1}.$$

By assumption, $\rho^{ab}(\mathrm{art}_k(t))$ acts trivially on the l-adic transcendental lattice $T_l(X_{\overline{k}})$. On the other hand, by Proposition 3.4, $\overline{\mathrm{Nm}_{k/E}(t)}/\mathrm{Nm}_{k/E}(t)$ also acts trivially. Hence $u(t) = 1$. By the definition of t, we have

$$(\overline{\mathrm{Nm}_{k/E}(t)}/\mathrm{Nm}_{k/E}(t))_\tau = 1,$$

and hence $\chi_\tau(t) = 1$.
$(b) \Rightarrow (a)$ Conversely, suppose that $\chi_\tau(t) = 1$. Then $u(t) = 1$. Let $x \in I_\mathfrak{p}$ be the image of t under the Artin map. It suffices to show that $\rho_l(x)$ acts trivially on $T_l(X_{\overline{k}})$. This follows from Proposition 3.4. □

Proposition 3.8. *Let $\mathfrak{p}$ be a prime ideal with $(l, \mathfrak{p}) = 1$, and let $\pi_\mathfrak{p} \in \mathscr{O}_\mathfrak{p}$ be a uniformizer. Suppose that ρ_l is unramified at $\mathfrak{p}$. Then the action of $u(\pi_\mathfrak{p}^{-1})$ on $T_l(X_{\overline{k}})$ coincides with the Frobenius action $\rho_l(\mathrm{Frob}_\mathfrak{p})$.*

Proof. The Artin map art_k sends $\pi_\mathfrak{p}^{-1}$ to $\mathrm{Frob}_\mathfrak{p}$. Hence we have

$$\rho(\mathrm{Frob}_\mathfrak{p}) = u(\pi_\mathfrak{p}^{-1})\overline{\mathrm{Nm}_{k/E}(\pi_\mathfrak{p}^{-1})}/\mathrm{Nm}_{k/E}(\pi_\mathfrak{p}^{-1}).$$

Taking the l-component, we obtain

$$\rho_l(\mathrm{Frob}_\mathfrak{p}) = u(\pi_\mathfrak{p}^{-1}).$$

□

Corollary 3.9. *Suppose that $(l, \mathfrak{p}) = 1$ and that ρ_l is unramified at $\mathfrak{p}$. Then the eigenvalues of $\mathrm{Frob}_\mathfrak{p}$ are given by*

$$\{\chi_\tau(\pi_\mathfrak{p}^{-1}) \mid \tau : E \to \mathbb{C} \text{ inclusions } \}.$$

In particular, the Galois representation ρ_l is rational.

Proof. Since X has CM over k, we have

$$\mathrm{rank}_{\mathbb{Z}} T(X_{\mathbb{C}}) = [E : \mathbb{Q}].$$

Hence the characteristic polynomial of $\mathrm{Frob}_{\mathfrak{p}}$ has degree $[E : \mathbb{Q}]$. By Proposition 3.8, the set

$$\{u(\pi_{\mathfrak{p}}^{-1})^{\tau} \mid \tau : E \to \mathbb{C} \textit{ inclusions } \}$$

gives all the eigenvalues of $\mathrm{Frob}_{\mathfrak{p}}$. On the other hand, we have

$$(\overline{\mathrm{Nm}_{k/E}(\pi_{\mathfrak{p}}^{-1})}/\mathrm{Nm}_{k/E}(\pi_{\mathfrak{p}}^{-1}))_{\tau} = 1.$$

Hence this set coincides with

$$\{\chi_{\tau}(\pi_{\mathfrak{p}}^{-1}) \mid \tau : E \to \mathbb{C} \}.$$

□

Remark 3.10. *The rationality is due to Dwork* [7]. *However, we prove this for CM K3 surfaces without Dwork's result.*

Corollary 3.11. *For each $\tau : E \to \mathbb{C}$, let $\rho_{l,\tau}$ be the one-dimensional l-adic representation associated with the Hecke character χ_{τ}. Then we have*

$$\rho_l \cong \bigoplus_{\tau : E \to \mathbb{C}} \rho_{l,\tau}.$$

Proof. This follows from Corollary 3.9, Remark 3.2, Chebotarev's density theorem. □

By Corollary 3.11, we obtain Theorem 1.3.

4. Examples; Kummer Surfaces associated to simple CM abelian surfaces

We now specialize Theorem 1.3 to Kummer surfaces associated to simple CM abelian surfaces. Using the classical CM theory of abelian varieties, we compare the Hecke characters obtained from Theorem 1.3 with those arising from the transcendental part of the abelian surface.

Following Chai-Conrad-Oort [2], we recall the definition of CM abelian varieties. Let K be a **CM algebra**, i.e., a finite product of CM fields. If $K = \prod K_i$, then we have $\mathrm{Hom}(K, \mathbb{C}) = \coprod_i \mathrm{Hom}(K_i, \mathbb{C})$. In particular, when $\dim_{\mathbb{Q}} K = 2g$, the set $\mathrm{Hom}(K, \mathbb{C})$ consists of exactly $2g$-elements. A **type** Φ of K is a subset $\Phi \subset \mathrm{Hom}(K, \mathbb{C})$ such that $\Phi \sqcup \overline{\Phi} = \mathrm{Hom}(K, \mathbb{C})$, where $\overline{(-)}$ denotes complex conjugation of $\mathbb{C}$. The pair (K, Φ) is called a **CM type**. If (K, Φ) is a CM type of K, then the $\mathbb{C}$-vector space $\bigoplus_{\sigma \in \Phi} \mathbb{C}$ has a canonical structure of a K-vector space via the diagonal map $K \ni x \mapsto (\sigma(x))_{\sigma \in \Phi} \in \bigoplus_{\sigma \in \Phi} \mathbb{C}$.

Definition 4.1. *(CM abelian varieties) Let K be a CM-algebra with $\dim_{\mathbb{Q}} K = g$, let (K, Φ) be a CM type, and let A be a g-dimensional abelian variety over a subfield $k \subset \mathbb{C}$. Suppose that an embedding $i : K \to \mathrm{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$ is given, and there is an isomorphism $T_0 A \otimes_k \mathbb{C} \cong \bigoplus_{\sigma \in \Phi} \mathbb{C}$ as $K \otimes_{\mathbb{Q}} \mathbb{C}$-modules, where $T_0 A$ is the tangent space of A at 0. The pair (A, i) is said to be a* ***CM abelian variety*** *over k of CM type (K, Φ).*

If (A, i) is a CM abelian variety over k of type (K, Φ), then k contains the reflex field

$$K^* := \mathbb{Q}(\sum_{\sigma \in \Phi} \sigma(x) | x \in K).$$

Indeed, by the definition of a CM abelian variety, there is a diagonal action $(\sigma(x))_{\sigma \in \Phi}$ on the k-vector space $T_0 A$ for every $x \in K$. Hence the trace $\sum_{\sigma \in \Phi} \sigma(x)$ lies in k.

Proposition 4.2. *Let (A, i) be a CM abelian surface over $k \subset \mathbb{C}$ of CM type (K, Φ). Then the Kummer surface $X := \mathrm{Km}(A)$ has CM over k by the reflex field $K^* \subset \mathbb{C}$.*

Proof. Let $\Phi = \{\sigma_1, \sigma_2\}$. By the construction of Kummer surfaces, there is a $\mathbb{Q}$-Hodge isometry (up to the double twist)

$$H^2(X_{\mathbb{C}}, \mathbb{Q}(1)) \cong H^2(A_{\mathbb{C}}, \mathbb{Q}(1)) \oplus \bigoplus_{i=1}^{16} [E_i],$$

where $[E_i]$ are the cohomology classes of the exceptional divisors of the minimal resolution $X \to A/\langle -1 \rangle$. In particular, we obtain a Hodge isometry (up to the double twist) $T(X_{\mathbb{C}})_{\mathbb{Q}} \cong T(A_{\mathbb{C}})_{\mathbb{Q}}$. Thus it suffices to prove that the reflex field of $T(A)_{\mathbb{Q}}$ is K^*.

Via the composition

$$K \xrightarrow{i} \mathrm{End}(A) \otimes \mathbb{Q} \to \mathrm{End}_{Hdg} T(A_{\mathbb{C}})_{\mathbb{Q}} \to \mathrm{End}_{\mathbb{C}} H^{2,0}(A_{\mathbb{C}}) = \mathbb{C}, \tag{4.1}$$

the field K acts on $H^{2,0}(A_{\mathbb{C}})$. For $x \in K$, the element $i(x)$ acts on $H^{2,0}(A)$ as $\sigma_1(x)\sigma_2(x)$. In fact, since $T_0(A_{\mathbb{C}}) \cong T_0 A \otimes_k \mathbb{C} \cong \bigoplus_{\sigma \in \Phi} \mathbb{C}$, the eigenvalues of $i(x)$ on $T_0(A_{\mathbb{C}})$ are $\sigma_1(x)$ and $\sigma_2(x)$. Hence the eigenvalue of $i(x)$ on $H^{2,0}(A_{\mathbb{C}}) = \wedge^2 H^{1,0}(A_{\mathbb{C}})$ is $\sigma_1(x)\sigma_2(x)$. Thus the image of the composition (4.1) is $K' := \mathbb{Q}(\sigma_1(x)\sigma_2(x) |\ x \in K) \subset \mathbb{C}$. On the other hand, K' coincides with the reflex field K^*. Indeed, we have the relations

$$\sigma_1(x) + \sigma_2(x) = \sigma_1(x+1)\sigma_2(x+1) - \sigma_1(x)\sigma_2(x) - 1,$$
$$2\sigma_1(x)\sigma_2(x) = (\sigma_1(x) + \sigma_2(x))^2 - (\sigma_1(x^2) + \sigma_2(x^2)),$$

which show that $K' = K^*$.

Let $E(T(A_{\mathbb{C}})_{\mathbb{Q}})$ denote the reflex field of $T(A_{\mathbb{C}})_{\mathbb{Q}}$. To show $K^* = E(T(A)_{\mathbb{Q}})$, we consider the following three cases.

(i) If $A_{\mathbb{C}}$ is simple, then the ring $\mathrm{End}(A_{\mathbb{C}})_{\mathbb{Q}}$ is a 4-dimensional CM field (see [2, 1.3.6.4. Proposition]). Hence K is a CM-field. By (4.1), the degree $[K^* : \mathbb{Q}]$ is equal to 4. Since the Picard number of $A_{\mathbb{C}}$ is 2 (see [19, Lemma 3.3]), we have $[E(T(A_{\mathbb{C}})_{\mathbb{Q}}) : \mathbb{Q}] = \dim_{\mathbb{Q}} T(A_{\mathbb{C}})_{\mathbb{Q}} = 4$, and hence $K^* = E(T(A)_{\mathbb{Q}})$.

(ii) If $A_{\mathbb{C}}$ is isogenous to a product of elliptic curves E_1 and E_2, and if E_1 and E_2 are not isogenous, then we have a canonical isomorphism of rings

$$\mathrm{End}(E_1)_{\mathbb{Q}} \times \mathrm{End}(E_2)_{\mathbb{Q}} \xrightarrow{\sim} \mathrm{End}(A_{\mathbb{C}})_{\mathbb{Q}}. \tag{4.2}$$

given by $(f_1, f_2) \mapsto f_1 \times f_2$. In fact, let $\varphi(x, y) = (f_1(x, y), f_2(x, y)) \in \mathrm{End}(A_{\mathbb{C}})_{\mathbb{Q}}$. Then

$$f_i(x, y) = f_i(x, 0) + f_i(0, y).$$

In particular, $f_1(0, y)$ is a homomorphism $E_2 \to E_1$. Since E_2 and E_1 are not isogenous, it follows that $f_1(0, y) = 0$ for all $y \in E_2$. By the same argument, $f_2(x, 0) = 0$ for all $x \in E_1$. Hence the morphism (4.2) is an isomorphism.

In particular, the CM algebra K is of the form $K_1 \times K_2$, where K_i are imaginary quadratic fields that are not isomorphic. Hence the reflex field K^* is generated by two non-isomorphic imaginary quadratic fields, and thus $[K^* : \mathbb{Q}] = 4$. On the other hand, since $\dim_{\mathbb{Q}} T(A)_{\mathbb{Q}} = 4$ (see [9, Corollary 2.3]), we obtain $[E(T(A)_{\mathbb{Q}}) : \mathbb{Q}] = 4$. Hence $K^* = E(T(A_{\mathbb{C}})_{\mathbb{Q}})$.

(iii) Suppose that $A_{\mathbb{C}}$ is isogenous to a product of two elliptic curves E_1 and E_2 which are isogenous. Then K is a product of two isomorphic imaginary quadratic fields. In particular, the reflex field K^* is a 2-dimensional vector space over $\mathbb{Q}$. Since the Picard number of $A_{\mathbb{C}}$ is 4 (see [9, Corollary 2.6]), we have $[E(T(A_{\mathbb{C}})_{\mathbb{Q}}) : \mathbb{Q}] = 2$. Thus $K^* = E(T(A)_{\mathbb{Q}})$. □

As a consequence of Milne's complex multiplication theory [17, Theorem 11.2] for abelian varieties, we will prove the main theorem of complex multiplication for the transcendental parts of simple abelian surfaces of CM type (Proposition 4.4).

Let (A, i) be a CM abelian surface over $\mathbb{C}$ of CM type (K, Φ). Via the embedding $i : K \hookrightarrow \mathrm{End}(A) \otimes \mathbb{Q}$, we see that the multiplicative group $K^\times \subset GL(H^1(A, \mathbb{Q}))$ is a maximal torus. Hence the Mumford-Tate group of $T(A)$ is contained in $K^\times$. In particular, the Hodge

structure $h : \mathbb{C}^\times \to GL(H^1(A,\mathbb{R}))$ factors through the torus $(K \otimes_{\mathbb{Q}} \mathbb{R})^\times$. We denote by μ_h the composition

$$\mathbb{C}^\times \ni t \mapsto (t,1) \in \mathbb{C}^\times \times \mathbb{C}^\times = \mathbb{S}(\mathbb{C}) \xrightarrow{h_{\mathbb{C}}} (K \otimes_{\mathbb{Q}} \mathbb{C})^\times = \prod_{\varphi \in \Phi} \mathbb{C}^\times \times \prod_{\varphi \in \overline{\Phi}} \mathbb{C}^\times$$

By the definition of h, we have $\mu_h(t) = (t^{-1}, t^{-1}, 1, 1) \in \prod_{\varphi\in\Phi} \mathbb{C}^\times \times \prod_{\varphi\in\overline{\Phi}} \mathbb{C}^\times$. Moreover, the minimal field of definition of μ_h is the reflex field K^*, since $\sigma\Phi = \Phi$ for every $\sigma \in \mathrm{Aut}(\mathbb{C}/K^*)$.

Proposition 4.3. *Let (A,i) be a CM abelian surface over $\mathbb{C}$ of type (K,Φ). Let $h : \mathbb{S} \to GL(H^1(A,\mathbb{R}))$ be the $\mathbb{Q}$-Hodge structure. The following statements hold:*
(1) The homomorphism

$$r_h^{-1} : \mathrm{Res}_{K^*/\mathbb{Q}}(\mathbb{G}_m) \xrightarrow{\mathrm{Res}_{K^*/\mathbb{Q}}(\mu_h^{-1})} \mathrm{Res}_{K^*/\mathbb{Q}}(\mathrm{Res}_{K/\mathbb{Q}}(\mathbb{G}_m)_{K^*}) \xrightarrow{\mathrm{Nm}_{K^*/\mathbb{Q}}} \mathrm{Res}_{K/\mathbb{Q}}(\mathbb{G}_m)$$

*is equal to the **reflex norm** N_Φ of CM-type (K,Φ) (see* [2, 2.1.3.3. Definition]*).*
(2) For any $x \in K^{*\times} \subset \mathbb{C}^\times$, we have

$$(\wedge^2\mu_h) \otimes \mu_{\mathbb{Q}(1)}(x) = \mu_{h^2}(x),$$

where $\wedge^2\mu_h$ is a cocharacter defined by $x \mapsto \mu_h(x) \wedge \mu_h(x) \in \wedge^2 H^1(A,\mathbb{C})$, μ_{h^2} is the cocharacter of the $\mathbb{Q}$-Hodge structure $H^2(A,\mathbb{Q})(1)$, and $\mu_{\mathbb{Q}(1)}$ is the cocharacter corresponding to the Tate twist $\mathbb{Q}(1)$.

Proof. (1) This follows from [2, 2.1.3.4. Proposition].
(2) This follows from the existence of a canonical $\mathbb{Q}$-Hodge isometry $H^2(A,\mathbb{Q})(1) \cong \wedge^2 H^1(A,\mathbb{Q}) \otimes \mathbb{Q}(1)$.

□

Proposition 4.4. *Let (A,i) be a CM simple abelian surface over $\mathbb{C}$ of type (K,Φ), let K^* be the reflex field of K, and let $\sigma \in \mathrm{Aut}(\mathbb{C}/K^*)$. Let $s \in \mathbb{A}_{K^*}^\times$ be an idéle such that $\mathrm{art}_{K^*}(s) = \sigma|_{K^{*ab}}$. Then there exists a K-linear isogeny $\alpha : A^\sigma \to A$ such that the following diagram commutes:*

$$\begin{array}{ccc} T(A)_{\mathbb{A}_f} & \xrightarrow{\alpha^*} & T(A^\sigma)_{\mathbb{A}_f} \\ {\scriptstyle \overline{s_f}/s_f}\uparrow & \nearrow_{\sigma^*} & \\ T(A)_{\mathbb{A}_f}, & & \end{array}$$

where $s_f \in \mathbb{A}_{K^*,f}^\times$ is the finite part of s.

Proof. For a prime l, let T_lA be the l-adic Tate module, let $T_fA := \prod_{l:prime} T_lA$, and let $V_fA := T_fA \otimes \mathbb{Q}$. By [17, Theorem 11.2], there exists a K-linear isogeny $\beta : A \to A^\sigma$ such that the following diagram commutes:

$$\begin{array}{ccc} V_fA & \xrightarrow{\beta} & V_fA^\sigma \\ {\scriptstyle N_\Phi(s)}\uparrow & \nearrow_{\sigma^{-1}} & \\ V_fA, & & \end{array}$$

where $\sigma^{-1} : A \to A^\sigma$ is the inverse of the conjugation $A^\sigma \to A$. Taking duals, we obtain the commutative diagram

$$\begin{array}{ccc} H^1(A,\mathbb{A}_f) & \xleftarrow{\beta^*} & H^1(A^\sigma,\mathbb{A}_f) \\ {\scriptstyle N_\Phi(s)}\downarrow & \swarrow_{\sigma^{-1*}} & \\ H^1(A,\mathbb{A}_f). & & \end{array}$$

Let $\alpha : A^\sigma \to A$ be the inverse isogeny of β. By Proposition 4.3 (1), the diagram

$$\begin{array}{ccc} H^1(A, \mathbb{A}_f) & \xrightarrow{\alpha^*} & H^1(A^\sigma, \mathbb{A}_f) \\ {\scriptstyle r_h(s)}\uparrow & \nearrow_{\sigma^*} & \\ H^1(A, \mathbb{A}_f). & & \end{array}$$

is commutative. By Proposition 4.3 (2), we obtain the following commutative diagram:

$$\begin{array}{ccc} H^2(A, \mathbb{A}_f)(1) & \xrightarrow{\alpha^*} & H^2(A^\sigma, \mathbb{A}_f)(1) \\ {\scriptstyle r_{h^2}(s)}\uparrow & \nearrow_{\sigma^*} & \\ H^2(A, \mathbb{A}_f)(1), & & \end{array}$$

where r_{h^2} is defined as the composition

$$\mathrm{Res}_{K^*/\mathbb{Q}}(\mathbb{G}_m) \xrightarrow{\mathrm{Res}_{K^*/\mathbb{Q}}(\mu_{h^2})} \mathrm{Res}_{K^*/\mathbb{Q}}(\mathrm{Res}_{K/\mathbb{Q}}(\mathbb{G}_m)_{K^*}) \xrightarrow{\mathrm{Nm}_{K^*/\mathbb{Q}}} \mathrm{Res}_{K/\mathbb{Q}}(\mathbb{G}_m).$$

Since $r_{h^2}(s)$ acts trivially on $\mathrm{NS}(A)_{\mathbb{Q}}$, it induces an action on $T(A)_{\mathbb{A}_f}$ via $\overline{s_f}/s_f$. Hence we obtain the required diagram. □

Corollary 4.5. *Let (A, i) be a simple CM abelian surface over a subfield $k \subset \mathbb{C}$ of CM-type (K, Φ). Let G be the Mumford-Tate group of $T(A_{\mathbb{C}})_{\mathbb{Q}}$. Then the Galois representation $\rho : \mathrm{Gal}(\overline{k}/k) \to GL(T(A_{\mathbb{C}})_{\mathbb{Q}})(\mathbb{A}_f)$ factors through a continuous homomorphism*

$$\rho^{ab} : \mathrm{Gal}(k^{ab}/k) \to G(\mathbb{A}_f)$$

such that the following diagram commutes:

$$\begin{array}{ccccc} \mathbb{A}_k^\times & \xrightarrow{\mathrm{art}_k} & \mathrm{Gal}(k^{ab}/k) & \xrightarrow{\rho^{ab}} & G(\mathbb{A}_f) \\ {\scriptstyle \mathrm{Nm}_{k/K^*}}\downarrow & & & & \downarrow \\ \mathbb{A}_{K^*}^\times & \xrightarrow[proj]{} & \mathbb{A}_{K^*,f}^\times & \xrightarrow[\overline{(-)}/(-)]{} & G(\mathbb{A}_f)/G(\mathbb{Q}). \end{array}$$

Proof. Our first step is to construct a continuous homomorphism $\rho : \mathrm{Gal}(k^{ab}/k) \to G(\mathbb{A}_f)$. Let $\sigma \in \mathrm{Gal}(\overline{k}/k)$. Choose an automorphism $\sigma' \in \mathrm{Aut}(\mathbb{C}/K^*)$ such that $\sigma'|_{\overline{k}} = \sigma$. Let $s \in \mathbb{A}_k^\times$ be an idéle such that $\mathrm{art}_k(s) = \sigma|_{k^{ab}}$. Then we have $\mathrm{art}_{K^*}(\mathrm{Nm}_{k/K^*}(s)) = \sigma|_{K^{*ab}}$. By Proposition 4.4, there exists a K-isogeny $\alpha : A_{\mathbb{C}} \to A_{\mathbb{C}}$ such that

$$\rho(\sigma) = \overline{\mathrm{Nm}_{k/K^*}(s)}/\mathrm{Nm}_{k/K^*}(s) \cdot \alpha^* \in G(\mathbb{A}_f). \tag{4.3}$$

Since the group $G(\mathbb{A}_f)$ is commutative, the Galois representation ρ is abelian; hence it induces a continuous homomorphism $\rho^{ab} : \mathrm{Gal}(k^{ab}/k) \to G(\mathbb{A}_f)$. Since $\alpha^* \in G(\mathbb{Q})$, it follows from (4.3) that the diagram is commutative. □

The following corollary is proved along the same vein as Theorem 1.3.

Corollary 4.6. *Let (A, i) be a CM simple abelian surface of type (K, Φ). Then there exists a unique locally constant homomorphism $u' : \mathbb{A}_k^\times \to K^{*\times}$ satisfying the following properties:*

(i) Fix an embedding $\tau : K^* \to \mathbb{C}$. The function

$$\chi'_\tau : \mathbb{A}_k^\times \xrightarrow{u'(-)\cdot\overline{\mathrm{Nm}_{k/K^*}(-)}/\mathrm{Nm}_{k/K^*}(-)} \mathbb{A}_{K^*}^\times \xrightarrow{\tau-\text{projection}} \mathbb{C}^\times$$

is an algebraic Hecke character.

(ii) Let

$$\rho_l : \mathrm{Gal}(\overline{k}/k) \to GL(T_l(A_{\overline{k}}))$$

be the l-adic representation. Then ρ_l is diagonalized by the algebraic Hecke characters χ_τ, where τ runs over the set $\mathrm{Hom}(K^*, \mathbb{C})$.

Proposition 4.7. *Let (A, i) be a CM simple abelian surface of type (K, Φ). Let $X := \mathrm{Km}(A)$ be the Kummer surface, and let χ_τ be the algebraic Hecke characters in Theorem 1.3 for X. Then, for every embedding $\tau \in \mathrm{Hom}(K^*, \mathbb{C})$, we have $\chi_\tau = \chi_\tau^{'}$.*

Proof. By the construction of Kummer surfaces, there is an isomorphism of $\mathrm{Gal}(\overline{k}/k)$-modules

$$H^2(X_{\overline{k}}, \mathbb{Q}_l)(1) \cong H^2(A_{\overline{k}}, \mathbb{Q}_l)(1) \oplus V,$$

where V is a 16-dimensional $\mathbb{Q}_l$-vector space and is generated by the cohomology classes of the exceptional divisors of the minimal resolution $X \to A/\langle -1 \rangle$. In particular, this induces an isomorphism of $\mathrm{Gal}(\overline{k}/k)$-modules

$$T(A_{\mathbb{C}})_{\mathbb{Q}_l} \cong T(X_{\mathbb{C}})_{\mathbb{Q}_l}.$$

For any $\sigma \in \mathrm{Gal}(\overline{k}/k)$, let $s \in \mathbb{A}_k^\times$ be an idéle such that $\sigma|_{k^{ab}} = \mathrm{art}_k(s)$. By Proposition 3.6, the action of σ on $T(X_{\mathbb{C}})_{\mathbb{Q}_l}$ is given by

$$u(t)\overline{\mathrm{Nm}_{k/K^*}(t)}/\mathrm{Nm}_{k/K^*}(t).$$

On the other hand, by Proposition 4.6, the action of σ on $T(A)_{\mathbb{Q}_l}$ is given by

$$u^{'}(t)\overline{\mathrm{Nm}_{k/K^*}(t)}/\mathrm{Nm}_{k/K^*}(t).$$

Hence we have

$$\chi_\tau(t) = \left(u(t)\frac{\overline{\mathrm{Nm}_{k/K^*}(t)}}{\mathrm{Nm}_{k/K^*}(t)} \right)_\tau = \left(u^{'}(t)\frac{\overline{\mathrm{Nm}_{k/K^*}(t)}}{\mathrm{Nm}_{k/K^*}(t)} \right)_\tau = \chi_\tau^{'}(t).$$

□

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